MIL-F-81100A(OS)
22 May 1980
SUPERSEDING
MIL-F-81100(OS)
24 June 1964
and AMENDMENT 3
8 September 1975

MILITARY SPECIFICATION

FLOAT, DRILL MINE, MARK 17 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Float, Drill Mine, Mark 17 Mod 0, LD 486283. The float is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-151	Metals; Test Methods
FED-STD-H28	Screw - Thread Standards for Federal Services

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501) Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment.
DOD-STD-2101	Classification of Characteristics

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

LD 486283	Float, Drill Mine, Mark 17 Mod 0
LD 496451	Container, Shipping, for Float Mark 17 Type

All drawings referenced herein are listed on LD 486283 unless otherwise designated.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. *REQUIREMENTS*

3.1 General requirements.

3.1.1 First article sample. Unless otherwise specified in the contract (see 6.2b), a first article sample of floats shall be manufactured and prepared for delivery using the methods and procedures proposed for the production lot, and tested for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing items that comply with this specification and the design. The sample size shall be as specified in the contract. The requirement for a first article may be waived by the contracting officer if satisfactory production has been recently accomplished by the contractor.

3.1.2 Fabrication. The floats shall be manufactured in accordance with the drawings listed on LD 486283 and as specified herein.

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3.1.3 Interchangeability. Component parts of assembled items (unless otherwise specified on the applicable drawings) fabricated in accordance with applicable drawings shall be interchangeable and shall assemble with mating components of the item without selection for fit or performance.

3.1.4 Screw threads. Unless otherwise specified, all screw threads shall be in accordance with FED-STD-H28. The class or fit of threads shall be as specified on the applicable drawing.

3.2 Performance requirements and product characteristics. The floats shall meet the following performance requirements and product characteristics:

3.2.1 Leakage. The Float Weldment, Dwg. 1874731, after the completion of all machining and grinding operations, and with all surfaces free of grease, paint, oil, or the like, shall withstand without leakage an internal pressure of $5\ ^{+1}_{-0}$ psig.

3.2.2 Seizing. The twine seizing securing the outer layer of the recovery line together, as illustrated on Dwg. 1874730, shall separate to free the recovery cable when subjected to a pull of between eight and twenty five pounds. At the first two points of securement, the pieces of twine shall remain attached to the nylon line after being broken.

3.2.3 Cable and loop pull requirement. Each end loop of the Nylon Recovery Cable, Dwg. 1800462; the Nylon Cable Assembly, Dwg. 1665784; or sample cable (FIGURE 1) shall withstand a 5800-pound minimum pull without breaking the cable, breaking the loop, or pulling the loop free.

3.3 Environmental requirements.

3.3.1 High pressure. The float weldment with flare well cover and inert Explosive Fitting Mark 20 Mod 0 in place shall withstand an external hydrostatic pressure of 67 ± 1 psig for a minimum of one hour without leakage or permanent deformation beyond the dimensional requirements of the drawings, except that the 1.839 ± 0.030-inch dimension to the flat surfaces on each side of the central spool slot shall not exceed $1.839\ ^{+0.280}_{-0.030}$ inches (see Dwg. 1874731).

3.3.2 Synthetic seawater spray. The float weldment shall withstand, without flaking of paint or corrosion of the base metal, the synthetic seawater spray test defined in FED-STD-151, Method 812.1, for a period of 48 hours.

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3.4 Clean-up. Prior to final assembly, all parts shall be clean; and protection against corrosion, dirt and other forms of contamination applied. The assembled float shall be free of burrs and sharp edges and, at time of packaging, shall be free of dirt, flaking or chipped paint, grease and oil (except as specified).

3.5 Marking. Each float assembly shall be identified by the marking "Float, Drill Mine, MK 17-0 Dwg. 486283" in accordance with MIL-STD-129.

3.6 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the floats shall be classified as follows:

a. First article inspection (see 4.5)

b. Quality conformance inspection (see 4.6)

c. Periodic production sample inspection (see 4.7)

4.3 Test equipment. The following items of test equipment are required to perform the test prescribed in this specification:

a. A source of regulated air pressure.

b. A well-lighted, water-filled tank large enough to permit submerging the float with room for rotating and observing all surfaces of the float. (The chamber of 4.3f may be used if it satisfies these requirements and the contractor is making all tests (see 6.2e).)

c. A rotatable mandrel capable of holding the float on the rotational axis.

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d. A pull scale capable of measuring a pull of 50 pounds in one-pound increments.

e. A pull tester capable of exerting a pull of 6000 ± 120 pounds on a nylon cable.

f. A test chamber capable of holding a Float Mark 17 Mod 0 and being pressurized to a hydrostatic pressure of 100 psig.

g. Necessary test apparatus to perform the synthetic seawater spray test of FED-STD-151, Method 812.1.

4.4 Test conditions. Unless otherwise specified, the float assemblies shall be subjected to tests under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F).

b. Altitude: Normal ground.

c. Humidity: Room ambient to 95 percent relative, maximum.

4.5 First article inspection.

4.5.1 First article testing. First article sample (see 3.1.1 and 6.2b) shall be subjected to the examinations and tests of 4.8 as specified in TABLE I and FIGURE 4 at the activity designated by the procuring agency (see 6.2e). Further production of the floats by the contractor, prior to approval of the first article, shall be at the contractor's risk.

4.5.2 First article approval. First article approval shall be granted when the Government, through the tests of FIGURE 4 and subsequent analysis of the test results, is assured of contractor conformance to the contract technical requirements. A first article may be disapproved when either nonconforming characteristics or unauthorized materials are found; however, conditional acceptance may be granted at the option of the Government based on analysis of detected defects.

4.6 Quality conformance inspection. Each float shall be subjected to the examinations and tests of 4.8 as designated in TABLE I. Failure of a unit to pass any examination or test shall be cause for rejection of the unit.

4.6.1 Sampling. See Note (3) under TABLE I.

4.7 Periodic production sample inspection. The periodic production sample of floats (see Note (1) under Table I) shall be randomly selected by the Government inspector from each production lot (see 6.2d) and subjected to the examinations and tests of 4.8 as designated in TABLE I and FIGURE 5 at the activity designated by the procuring agency (see 6.2e). A unit failing any designated periodic sample test may cause the sample and the lot it represents to be rejected.

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4.8 Examination and test methods.

4.8.1 Packaging. The marking and packing shall be examined by the contractor prior to shipment and by the Government upon receipt to determine conformance to 3.5 and Section 5.

4.8.2 Examination.

4.8.2.1 Cleanliness. Prior to final assembly, verify that all parts are cleaned and protected against corrosion, dirt, and other forms of contamination. After assembly, verify that the float is free of burrs and sharp edges. At the time of packing, verify that the float is free of dirt, flaking or chipped paint, and oil or grease other than where specified.

4.8.2.2 Fabrication. Examine the float assemblies in accordance with Dwg. 1874730 to determine compliance with this specification and the drawings as specified in 3.1.2. Examination in accordance with Dwg. 1874730 shall not relieve the contractor of his responsibility to meet all requirements of other drawings listed or referenced thereon.

4.8.3 Leakage (M101). Remove the Self-Sealing Bolt, Dwg. 2430629-18, from the base of the flare-well member of the Float Weldment, Dwg. 1874731, and connect in its place the source of regulated air pressure (4.3a). Apply internally to the weldment an air pressure of 5 $^{+1}_{-0}$ psig. While maintaining this pressure, submerge the weldment in water (4.3b) and brush away all air bubbles. Under adequate lighting, examine all surfaces including the flare-well cavity. Verify that bubbles indicating leakage are not escaping from the weldment.

4.8.4 Seizing (sampling) (M102). Attach a suitable, freely rotatable mandrel with its axis of rotation perpendicular to the horizontal surface of any stationary object. Position the float assembly on the mandrel (4.3c), utilizing the float's signal well. With the scale (4.3d),grip the payout end of the recovery cable twelve inches from the float and steadily pull the cable outward, radially to the axis of rotation of the float, until the cotton thread seizing securing the outer loops together breaks and frees the cable. Regrip the recovery cable twelve inches from the next point of seizing. Pull until the seizing breaks and releases the cable. Verify that the force required to separate these two points of securement is not less than eight pounds or more than twenty five pounds, and that the broken ends of the first point of securement remain attached to the nylon cable. Continue this procedure until all loops comprising the top layer of cable about the float have been released. For these remaining loops, verify that the force required to free the cable is not more than twenty five pounds.

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4.8.5 Cable and loop pull tests.

4.8.5.1 Nylon recovery cable proofing (sampling) (M103). With the Nylon Recovery Cable, Dwg. 1800462, removed from the float assembly, secure the cable loop to the pull arm of the tester (4.3e) in the manner shown in FIGURE 2. Pass at least five turns of the cable about a fixed support and secure as specified. Apply a steadily increasing load until the cable fails. Verify that the cable did not break and that the loops did not pull out or break under a load of less than 5800 pounds.

4.8.5.2 Nylon cable assembly proofing (sampling) (M104). Remove the Nylon Cable Assembly, Dwg. 1665784, from the float assembly and secure it to an appropriate fixture on the tester (4.3e) as shown in FIGURE 3. Apply a steadily increasing load until the cable fails. Verify that the cable did not break and that the loops did not pull out or break under a load of less than 5800 pounds.

4.8.5.3 Sample cable proofing (M105). Four sample cables as shown in FIGURE 1 shall be fabricated and tested prior to initiating production. The sample cables shall be made from the same nylon cable that is used to make the Nylon Recovery Cable, Dwg. 1800462, and the Nylon Cable Assembly, Dwg. 1665784. The splicing techniques used in making the sample cable loops shall be in accordance with the requirements of Dwg. 1665784 and 1800462. Attach each end of the sample cable to the pull arms of the tester (4.3e) with ½-inch diameter pins. Apply a steadily increading load until the cable fails. Verify that the cable did not break and that the loops did not pull out or break under a load of less than 5800 pounds.

4.9 Environmental tests.

4.9.1 High pressure. Install and secure the flare well cover on the float as in service use. Install suitable watertight plugs in the holes in the base of the signal tube. Weigh the dry float to an accuracy of one ounce and record the weight. Place the float in the test chamber (4.3f) and pressurize the chamber to 67 ± 1 psig hydrostatic pressure. Keep the chamber pressurized for one hour minimum. Reduce the pressure in the chamber to zero psig and remove the float from the chamber. Carefully dry all external surfaces of the float. Again, weigh the float. Verify that the float has not increased in weight over the above recorded weight of the dry float. Verify that there is no water in the flare well and that the dimensions of the flare well and float conform to the appropriate drawings. In addition, verify that the dimension to the flat surfaces on each side of the central spool slot does not exceed $1.839\ ^{+0.280}_{-0.030}$ inches.

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4.9.2 Synthetic seawater spray test (M106). Mount the painted, completely assembled float, including the flare well cover, less recovery cable and cable assembly, in the test apparatus required in 4.3g. Subject the float to the synthetic seawater spray test defined in FED-STD-151, Method 812.1, for a period of 48 hours. After completing the test, rinse the float in clean water, and dry the float. Verify that there is no evidence of flaking paint or corrosion of the base metal.

4.10 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as major or minor in accordance with DOD-STD-2101. Tests and examinations that verify major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the tests or examination. Tests and examinations which are not annotated with a classification code are classified minor. (See Note (3) under TABLE I.

5. PACKAGING

5.1 Preservation-packaging. (Not applicable)

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each float assembly shall be individually packed in accordance with LD 496451.

5.3 Markings.

5.3.1 Special markings. (Not applicable)

5.3.2 Normal markings. In addition to the markings required by contract, the shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and 3.5.

6. NOTES

6.1 Intended use. The Float Mark 17 Mod 0 is intended for use with the Actuation Dummy Mines Marks 52 and 55. The Float Mark 17 Mod 0 is a bouyant device which has a tube for containing a smoke signal and facilities for ejecting the signal when proper external influences are received. In addition, the float is configured to provide space for a nylon recovery cable.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Number of first articles required (see 3.1.1 and 4.5).

c. Number of items in the periodic production sample (see 4.7).

d. Lot size for sampling purposes, if one month's scheduled production is not acceptable (see 4.7 and 6.3.2). Lot shall not exceed 25 items.

e. Activity to perform first article and periodic production sample inspection, if other than the contractor (see 4.5 and 4.7).

6.3 Definitions. The following terms are used throughout this specification and are here defined for clarity.

6.3.1 Float. Reference herein to the float unit, item or assembly shall be understood to mean the Float, Drill Mine Mark 17 Mod 0 Assembly.

6.3.2 Lot. Unless otherwise specified in the contract, lot size shall mean one month's scheduled production. See Section 5 of MIL-STD-105 for lot composition.

6.4 The design and technical cognizance for the float is located at the Naval Mine Engineering Facility, Yorktown, Virginia 23691.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE I. Correlation of requirements and test methods.
(see Notes at end of table)

Title of Examination or Test	Requirement Paragraph	Method Paragraph	Inspection Classifications (1) First (2) Article	Quality (3) Conformance	Periodic (2) Production
Group A					
Packaging Delivery	Section 5	4.8.1	X	X	X
Visual	3.1.2, 3.1.3 3.1.4, 3.4 and 3.6	4.8.2.1 and 4.8.2.2	X	X	X
Leakage	3.2.1	4.8.3	X(4)	X	X(4)
Seizing	3.2.2	4.8.4	X		X
Nylon recovery cable proofing	3.2.3	4.8.5.1	X		X
Nylon cable assembly proofing	3.2.3	4.8.5.2	X		X
Sample cable proofing	3.2.3	4.8.5.3	X(5)		
Group B					
High pressure	3.3.1	4.9.1	X		X
Synthetic seawater spray	3.3.2	4.9.2	X		X

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TABLE I. Correlation of requirements and test methods. - Continued

NOTES

(1) Unless otherwise specified in the procurement document, inspection shall be in accordance with MIL-STD-105, Inspection Level II, and acceptance quality levels (AQL's) of 2.5% and 10% for major and minor defect classification, respectively.

(2) The sequence for performing the inspection of the first article and periodic production samples shall be determined by the designated Government testing activity.

(3) Quality conformance inspection shall be an in-process function performed during fabrication of the float prior to preparation for delivery by the manufacturer.

(4) This test may be performed on the painted first article and periodic samples at the discretion of the designated Government testing activity.

(5) Fabricate (and test) in accordance with 4.8.5.3.

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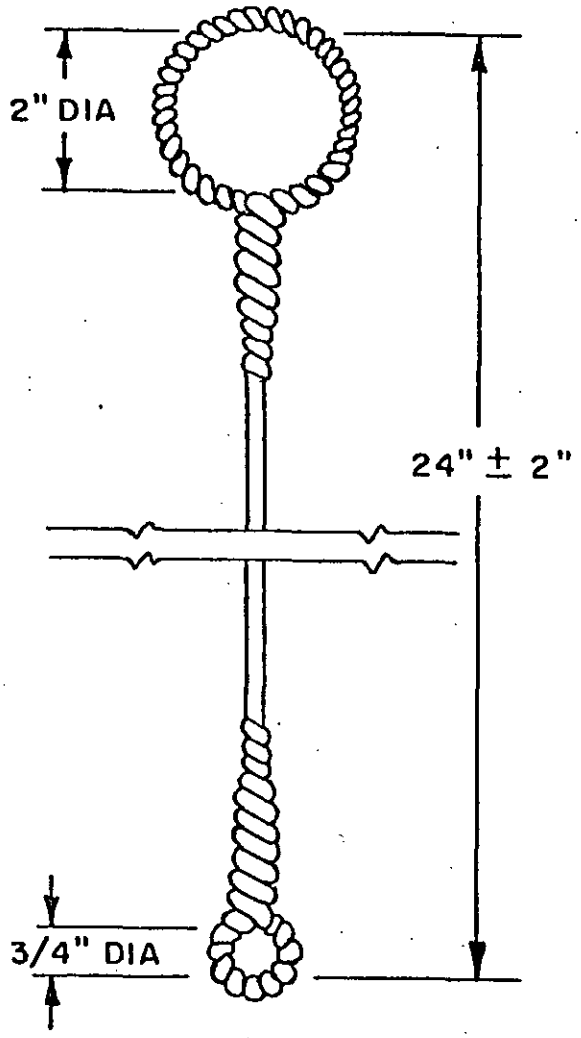


FIGURE 1. Sample cable for cable loop pull tests.

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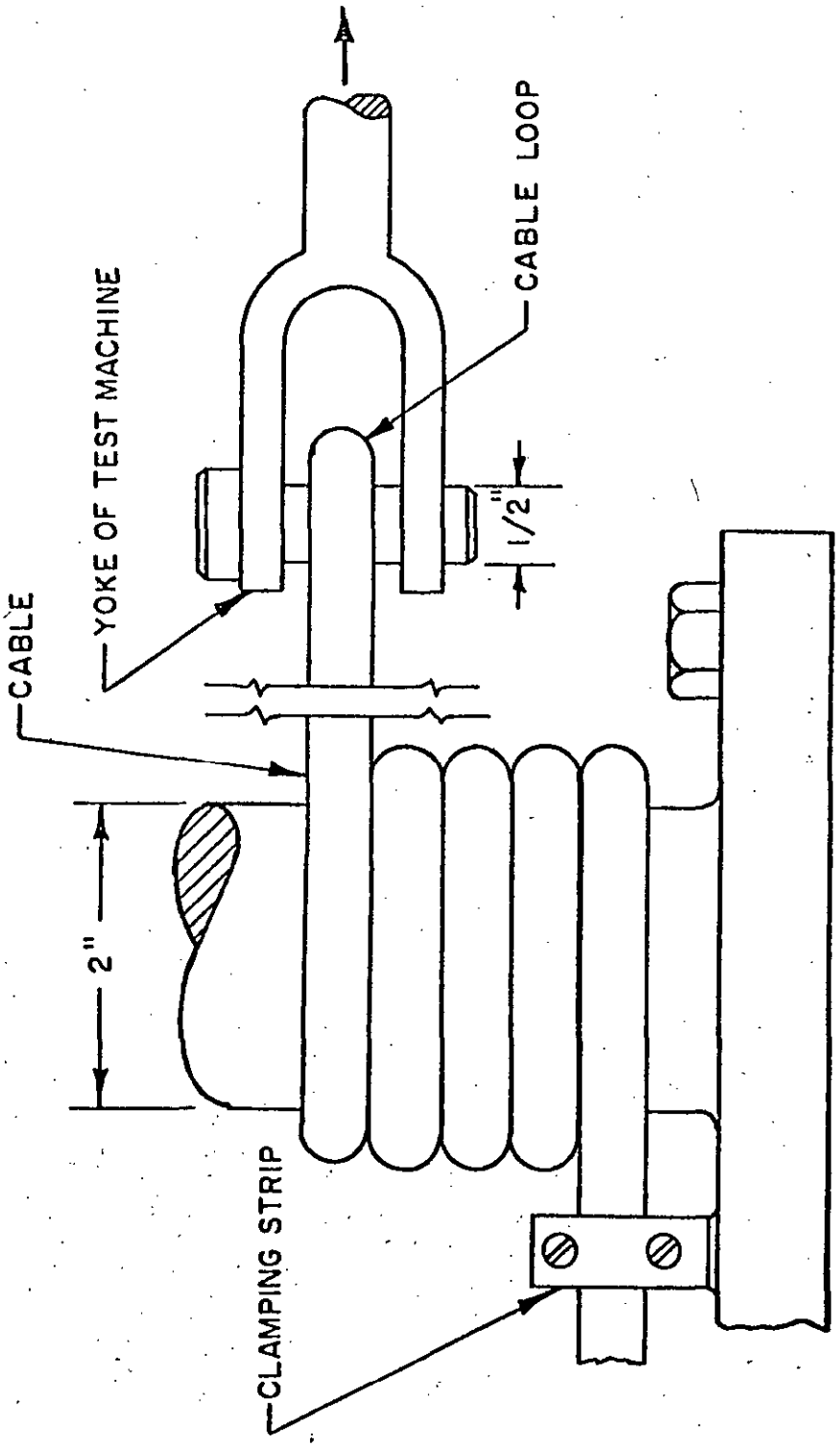


FIGURE 2. Cable loop pull test for the recovery cable (Dwg. 1800462).

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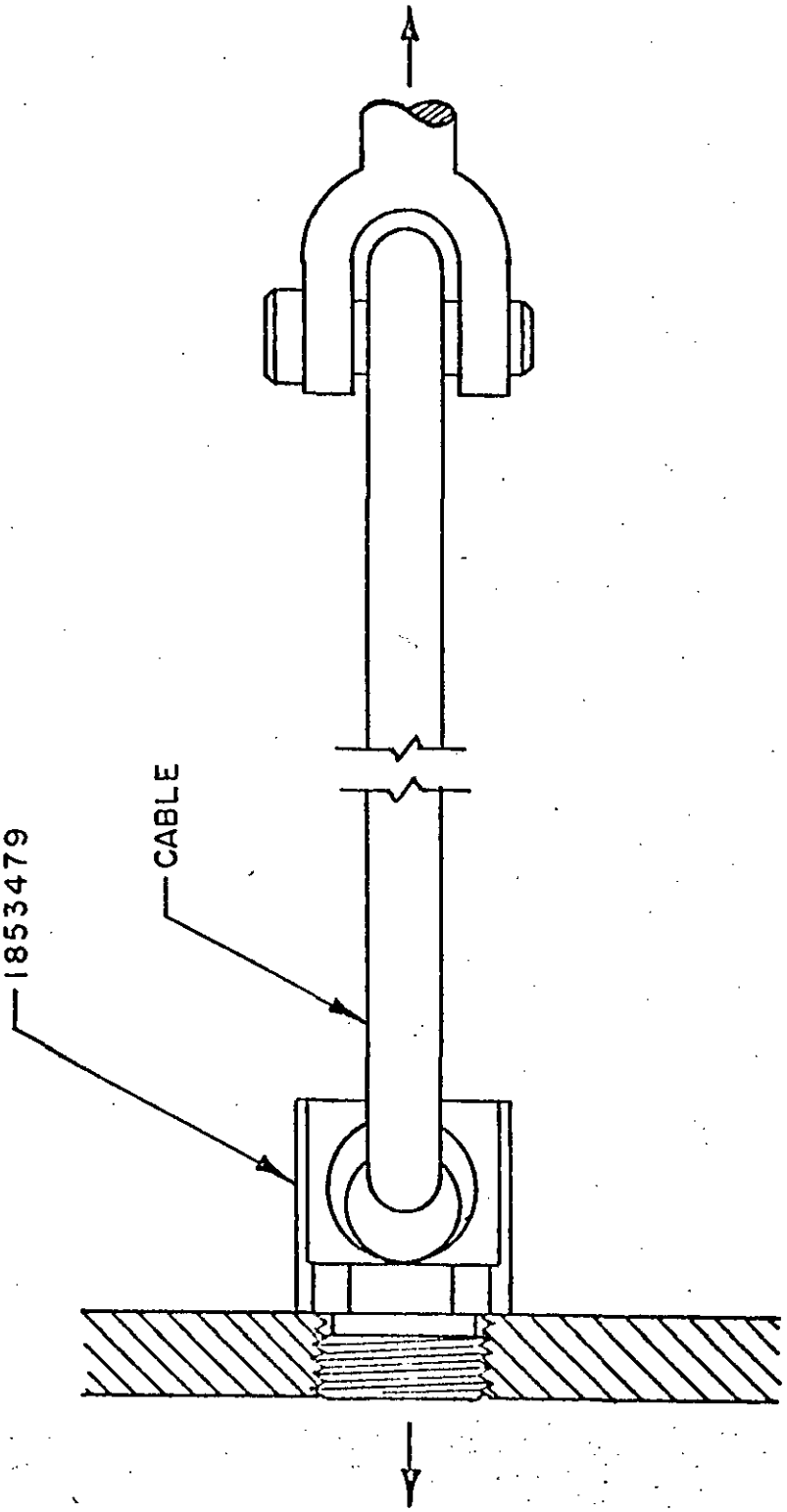
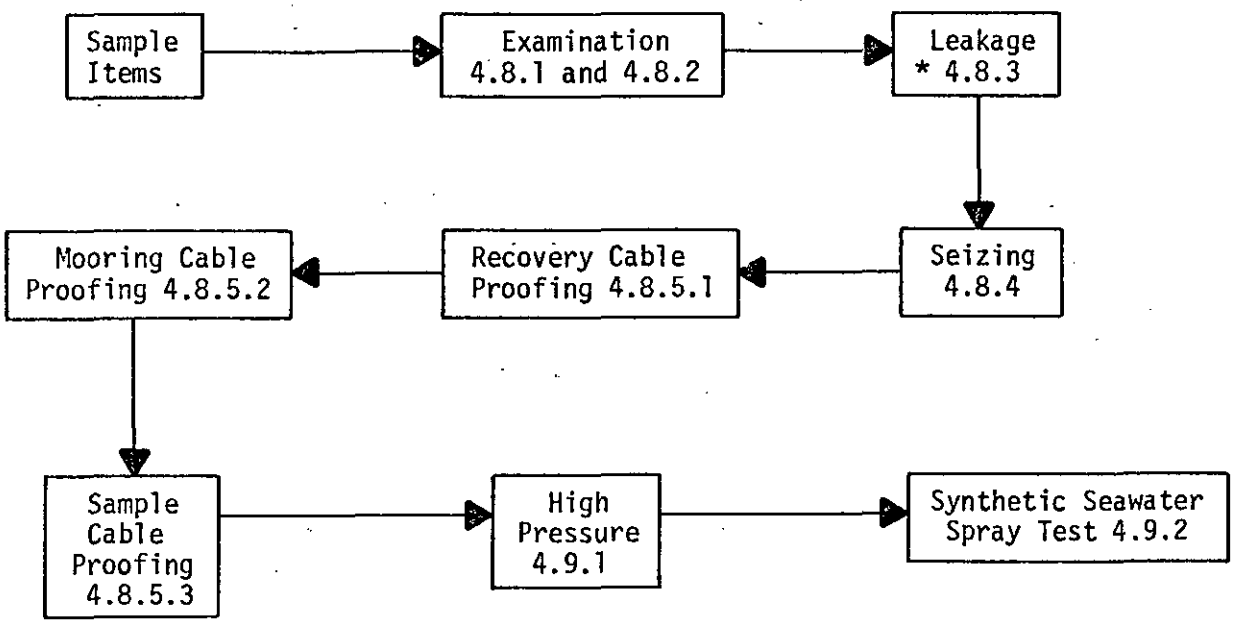


FIGURE 3. Cable loop pull test for the mooring cable (Dwg. 1665784).

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Drill Mine Float Mark 17 Mod 0

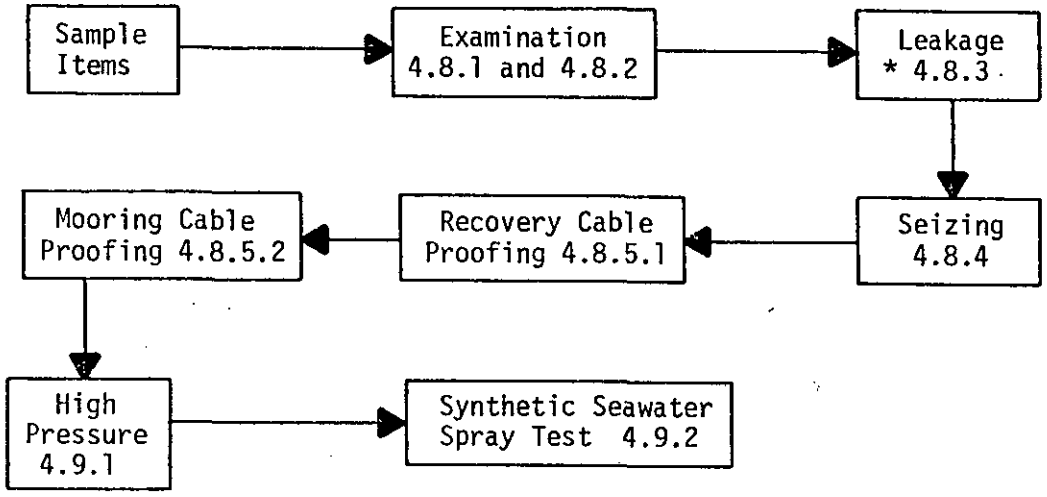


*See Note (4) under TABLE I.

FIGURE 4. First article sample tests.

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Drill Mine Float Mark 17 Mod 0



*See Note (4) under TABLE I.

FIGURE 5. Periodic production sample tests.

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-81100A "FLOAT, DRILL MINE, MARK 17 MOD 0"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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